

Name - Ajeem Aniskhan Makandar
PRN - 240850320007

1. List all the columns of the Salespeople table.

```
mysql> Describe Salespeople;
```

Field	Type	Null	Key	Default	Extra
snum	int	NO	PRI	NULL	
sname	varchar(20)	YES		NULL	
city	varchar(20)	YES		NULL	
comm	float	YES		NULL	

2. List all customers with a rating of 100.

```
mysql> select * from customers where rating = 100;
```

cnum	cname	city	rating	snum
2001	hoffman	london	100	1001
2006	clemens	london	100	1001
2007	pareira	rome	100	1004

3. Find all records in the Customer table with NULL values in the city column

```
mysql> SELECT * FROM customers WHERE CITY IS NULL;  
Empty set (0.00 sec)
```

4. Find the largest order taken by each salesperson on each date.

```
SELECT SNUM, ODATE, MAX(AMT) AS max_amount FROM  
-> orders GROUP BY SNUM, ODATE;
```

SNUM	ODATE	max_amount
1007	1996-03-10	1098.16
1004	1996-03-10	1900.1
1001	1996-03-10	767.19
1002	1996-03-10	5160.45
1003	1996-04-10	1713.23
1001	1996-05-10	4723
1002	1996-06-10	1309.95
1001	1996-06-10	9891.88

5. Arrange the Orders table by descending customer number.

```
mysql> SELECT * FROM orders ORDER BY CNUM DESC;
```

ONUM	AMT	ODATE	CNUM	SNUM
3001	18.69	1996-03-10	2008	1007
3006	1098.16	1996-03-10	2008	1007
3002	1900.1	1996-03-10	2007	1004
3008	4723	1996-05-10	2006	1001
3011	9891.88	1996-06-10	2006	1001
3010	1309.95	1996-06-10	2004	1002
3005	5160.45	1996-03-10	2003	1002
3007	75.75	1996-04-10	2002	1003
3009	1713.23	1996-04-10	2002	1003
3003	767.19	1996-03-10	2001	1001

6. Find which salespeople currently have orders in the Orders table.

```
mysql> SELECT DISTINCT s.SNUM, s.SNAME FROM salespeople s JOIN  
-> orders o ON s.SNUM = o.SNUM;
```

SNUM	SNAME
1001	peel
1002	serres
1003	axelrod
1004	motika
1007	rifkin

7. List names of all customers matched with the salespeople serving them.

```
mysql> SELECT c.CNAME, s.SNAME FROM customers c JOIN salespeople s  
-> ON c.SNUM = s.SNUM;
```

CNAME	SNAME
hoffman	peel
givonanni	axelrod
liu	serres
grass	serres
clemens	peel
pareira	motika
cisneros	rifkin

8. Find the names and numbers of all salespeople who had more than one customer.

```
SELECT s.SNUM, s.SNAME FROM salespeople s JOIN customers c
-> ON s.SNUM = c.SNUM GROUP BY s.SNUM, s.SNAME HAVING
-> COUNT(c.CNUM) > 1;
```

SNUM	SNAME
1001	peel
1002	serres

9.Count the orders of each of the salespeople and output the results in descending order.

```
mysql> SELECT SNUM, COUNT(ONUM) AS order_count FROM orders
-> GROUP BY SNUM ORDER BY order_count DESC;
```

SNUM	order_count
1001	3
1002	2
1003	2
1007	2
1004	1

10.List the Customer table if and only if one or more of the customers in the Customer table are located in San Jose.

```
mysql> SELECT * FROM customers WHERE EXISTS (SELECT 1 FROM
-> customers WHERE CITY = 'San Jose');
```

cnum	cname	city	rating	snum
2001	hoffman	london	100	1001
2002	givonanni	rome	200	1003
2003	liu	san jose	200	1002
2004	grass	berlin	300	1002
2006	clemens	london	100	1001
2007	pareira	rome	100	1004
2008	cisneros	san jose	300	1007

11.Match salespeople to customers according to what city they lived in.

```
mysql> SELECT s.SNAME, c.CNAME, s.CITY FROM salespeople s JOIN  
-> customers c ON s.CITY = c.CITY;
```

SNAME	CNAME	CITY
fran	hoffman	london
motika	hoffman	london
peel	hoffman	london
serres	liu	san jose
fran	clemens	london
motika	clemens	london
peel	clemens	london
serres	cisneros	san jose

12.Find the largest order taken by each salesperson.

```
mysql> SELECT SNUM, MAX(AMT) AS largest_order FROM orders GROUP  
-> BY SNUM;
```

SNUM	largest_order
1001	9891.88
1002	5160.45
1003	1713.23
1004	1900.1
1007	1098.16

13.Find customers in San Jose who have a rating above 200.

```
mysql> SELECT * FROM customers WHERE CITY = 'San Jose' AND  
-> RATING > 200;
```

cnum	cname	city	rating	snum
2008	cisneros	san jose	300	1007

14.List the names and commissions of all salespeople in London.

```
mysql> SELECT SNAME, COMM FROM salespeople WHERE CITY =  
-> 'London';
```

SNAME	COMM
peel	0.12
motika	0.11
fran	0.26

15. List all the orders of salesperson Motika from the Orders table.

```
SELECT * FROM orders WHERE SNUM = ( SELECT SNUM FROM
-> salespeople WHERE SNAME = 'Motika');
```

ONUM	AMT	ODATE	CNUM	SNUM
3002	1900.1	1996-03-10	2007	1004

16. Find all customers with orders on October 3.

```
mysql> SELECT DISTINCT c.* FROM customers c JOIN orders o ON c.CNUM =
-> o.CNUM WHERE o.ODATE = '1996-03-10';
```

cnum	cname	city	rating	snum
2008	cisneros	san jose	300	1007
2007	pareira	rome	100	1004
2001	hoffman	london	100	1001
2003	liu	san jose	200	1002

17. Give the sums of the amounts from the Orders table, grouped by date, eliminating all those dates where the SUM was not at least 2000.00 above the MAX amount.

```
mysql> select odate,sum(amt) from orders group by odate having
-> sum(amt)<(2000+max(amt));
```

odate	sum(amt)
1996-04-10	1788.97998046875
1996-05-10	4723
1996-06-10	11201.829833984375

18. Select all orders that had amounts that were greater than at least one of the orders from October 6.

```
mysql> SELECT * FROM orders WHERE AMT > ANY ( SELECT AMT
-> FROM orders WHERE ODATE = '1996-06-10');
```

ONUM	AMT	ODATE	CNUM	SNUM
3002	1900.1	1996-03-10	2007	1004
3005	5160.45	1996-03-10	2003	1002
3008	4723	1996-05-10	2006	1001
3009	1713.23	1996-04-10	2002	1003
3011	9891.88	1996-06-10	2006	1001

19. Write a query that uses the EXISTS operator to extract all salespeople who have customers with a rating of 300.

```
mysql> SELECT DISTINCT s.SNUM, s.SNAME FROM salespeople s
-> WHERE EXISTS ( SELECT 1 FROM customers c WHERE c.SNUM =
-> s.SNUM AND c.RATING = 300);
```

```
+-----+-----+
| SNUM | SNAME |
+-----+-----+
| 1002 | serres |
| 1007 | rifkin |
+-----+-----+
```

20. Find all pairs of customers having the same rating.

```
mysql> SELECT c1.CNAME AS Customer1, c2.CNAME AS Customer2, c1.RATING
-> FROM customers c1 JOIN customers c2 ON c1.RATING = c2.RATING AND
-> c1.CNUM <> c2.CNUM;
```

```
+-----+-----+-----+
| Customer1 | Customer2 | RATING |
+-----+-----+-----+
| pareira | hoffman | 100 |
| clemens | hoffman | 100 |
| liu | givonanni | 200 |
| givonanni | liu | 200 |
| cisneros | grass | 300 |
| pareira | clemens | 100 |
| hoffman | clemens | 100 |
| clemens | pareira | 100 |
| hoffman | pareira | 100 |
| grass | cisneros | 300 |
+-----+-----+-----+
```

21. Find all customers whose CNUM is 1000 above the SNUM of Serres.

```
mysql> SELECT * FROM customers WHERE CNUM = (SELECT SNUM +
-> 1000 FROM salespeople WHERE SNAME = 'Serres');
```

```
+-----+-----+-----+-----+-----+
| cnum | cname | city | rating | snum |
+-----+-----+-----+-----+-----+
| 2002 | givonanni | rome | 200 | 1003 |
+-----+-----+-----+-----+-----+
```

22. Give the salespeople's commissions as percentages instead of decimal numbers.

```
mysql> SELECT SNUM, SNAME, COMM * 100 AS COMM_PERCENTAGE  
-> FROM salespeople;
```

SNUM	SNAME	COMM_PERCENTAGE
1001	peel	11.999999731779099
1002	serres	12.999999523162842
1003	axelrod	10.000000149011612
1004	motika	10.999999940395355
1005	fran	25.999999046325684
1007	rifkin	15.000000596046448

23.. Find the largest order taken by each salesperson on each date, eliminating those MAX orders which are less than \$3000.00 in value.

```
mysql> WITH MaxOrders AS ( SELECT SNUM, ODATE, MAX(AMT) AS  
-> max_amount FROM orders GROUP BY SNUM, ODATE) SELECT * FROM  
-> MaxOrders WHERE max_amount >= 3000.00;
```

SNUM	ODATE	max_amount
1002	1996-03-10	5160.45
1001	1996-05-10	4723
1001	1996-06-10	9891.88

24. List the largest orders for October 3, for each salesperson.

```
mysql> WITH October3Orders AS (SELECT SNUM, MAX(AMT) AS max_amount  
FROM orders WHERE ODATE = '1996-03-10' GROUP BY SNUM) SELECT o.SNUM,  
o.ODATE, o.AMT FROM orders o JOIN October3Orders o3 ON o.SNUM = o3.SNUM  
AND o.AMT = o3.max_amount;
```

SNUM	ODATE	AMT
1001	1996-03-10	767.19
1002	1996-03-10	5160.45
1004	1996-03-10	1900.1
1007	1996-03-10	1098.16

25. Find all customers located in cities where Serres (SNUM 1002) has customers.

```
mysql> SELECT DISTINCT c.* FROM customers c WHERE CITY IN  
-> (SELECT CITY FROM customers WHERE SNUM = 1002);
```

cnum	cname	city	rating	snum
2003	liu	san jose	200	1002
2004	grass	berlin	300	1002
2008	cisneros	san jose	300	1007

26. Select all customers with a rating above 200.00.

```
mysql> SELECT * FROM customers WHERE RATING > 200;
```

cnum	cname	city	rating	snum
2004	grass	berlin	300	1002
2008	cisneros	san jose	300	1007

27. Count the number of salespeople currently listing orders in the Orders table.

```
mysql> SELECT COUNT(DISTINCT SNUM) AS salesperson_count FROM orders;
```

salesperson_count
5

Q28. Write a query that produces all customers serviced by salespeople with a commission above 12%. Output the customer's name and the salesperson's rate of commission.

```
mysql> SELECT c.CNAME, s.COMM FROM customers c JOIN salespeople s ON  
c.SNUM = s.SNUM WHERE s.COMM > 0.12;
```

CNAME	COMM
liu	0.13
grass	0.13
cisneros	0.15

29. Find salespeople who have multiple customers.

```
mysql> SELECT s.SNUM, s.SNAME FROM salespeople s JOIN customers c ON  
s.SNUM = c.SNUM GROUP BY s.SNUM, s.SNAME HAVING COUNT(c.CNUM) > 1;
```

```
+-----+-----+  
| SNUM | SNAME |  
+-----+-----+  
| 1001 | peel  |  
| 1002 | serres|  
+-----+-----+
```

30. Find salespeople with customers located in their city.

```
mysql> SELECT s.SNUM, s.SNAME FROM salespeople s JOIN customers c ON  
s.SNUM = c.SNUM GROUP BY s.SNUM, s.SNAME HAVING COUNT(c.CNUM) > 1;
```

```
+-----+-----+  
| SNUM | SNAME |  
+-----+-----+  
| 1001 | peel  |  
| 1002 | serres|  
+-----+-----+
```

31. Find all salespeople whose name starts with 'P' and the fourth character is 'l'.

```
mysql> SELECT SNUM, SNAME FROM salespeople WHERE SNAME LIKE 'P__l%';
```

```
+-----+-----+  
| SNUM | SNAME |  
+-----+-----+  
| 1001 | peel  |  
+-----+-----+
```

32. Write a query that uses a subquery to obtain all orders for the customer named Cisneros. Assume you do not know his customer number.

```
mysql> SELECT * FROM orders WHERE CNUM = ( SELECT CNUM FROM  
customers WHERE CNAME = 'Cisneros');
```

```
+-----+-----+-----+-----+-----+  
| ONUM | AMT   | ODATE   | CNUM | SNUM |  
+-----+-----+-----+-----+-----+  
| 3001 | 18.69 | 1996-03-10 | 2008 | 1007 |  
| 3006 | 1098.16 | 1996-03-10 | 2008 | 1007 |  
+-----+-----+-----+-----+-----+
```

33. Find the largest orders for Serres and Rifkin.

```
mysql> SELECT SNAME, MAX(AMT) AS largest_order FROM orders o JOIN  
salespeople s ON o.SNUM = s.SNUM WHERE s.SNAME IN ('Serres', 'Rifkin') GROUP  
BY SNAME;
```

SNAME	largest_order
serres	5160.45
rifkin	1098.16

34. Extract the Salespeople table in the following order : SNUM, SNAME,
COMMISSION, CITY.

```
mysql> SELECT SNUM, SNAME, COMM AS COMMISSION, CITY FROM  
salespeople;
```

SNUM	SNAME	COMMISSION	CITY
1001	peel	0.12	london
1002	serres	0.13	san jose
1003	axelrod	0.1	new york
1004	motika	0.11	london
1005	fran	0.26	london
1007	rifkin	0.15	barcelona

35. Select all customers whose names fall in between 'A' and 'G' alphabetical range.

```
mysql> SELECT * FROM customers WHERE CNAME BETWEEN 'A' AND 'G';
```

cnum	cname	city	rating	snum
2006	clemens	london	100	1001
2008	cisneros	san jose	300	1007

Q36. Select all the possible combinations of customers that you can assign.

```
mysql> SELECT c1.CNAME AS Customer1, c2.CNAME AS Customer2 FROM
customers c1, customers c2 WHERE c1.CNUM <> c2.CNUM;
```

```
+-----+-----+
| Customer1 | Customer2 |
+-----+-----+
| cisneros  | hoffman   |
| pareira   | hoffman   |
| clemens   | hoffman   |
| grass     | hoffman   |
| liu       | hoffman   |
| givonanni | hoffman   |
| cisneros  | givonanni |
| pareira   | givonanni |
| clemens   | givonanni |
| grass     | givonanni |
| liu       | givonanni |
| hoffman   | givonanni |
| cisneros  | liu       |
| pareira   | liu       |
| clemens   | liu       |
| grass     | liu       |
| givonanni | liu       |
| hoffman   | liu       |
| cisneros  | grass     |
| pareira   | grass     |
| clemens   | grass     |
| liu       | grass     |
| givonanni | grass     |
| hoffman   | grass     |
| cisneros  | clemens   |
| pareira   | clemens   |
| grass     | clemens   |
| liu       | clemens   |
| givonanni | clemens   |
| hoffman   | clemens   |
| cisneros  | pareira   |
| clemens   | pareira   |
| grass     | pareira   |
| liu       | pareira   |
| givonanni | pareira   |
| hoffman   | pareira   |
| pareira   | cisneros  |
| clemens   | cisneros  |
| grass     | cisneros  |
| liu       | cisneros  |
| givonanni | cisneros  |
| hoffman   | cisneros  |
+-----+-----+
```

37. Select all orders that are greater than the average for October 4.

```
mysql> WITH October4Orders AS (SELECT AMT FROM orders WHERE ODATE =
'1996-04-10'), AverageAmount AS ( SELECT AVG(AMT) AS avg_amount FROM
October4Orders)SELECT * FROM orders WHERE ODATE = '1996-04-10' AND AMT
> (SELECT avg_amount FROM AverageAmount);
```

+	-----	+	-----	+	-----	+	-----	+		
	ONUM		AMT		ODATE		CNUM		SNUM	
+	-----	+	-----	+	-----	+	-----	+	-----	+
	3009		1713.23		1996-04-10		2002		1003	
+	-----	+	-----	+	-----	+	-----	+	-----	+

38. Write a select command using a correlated subquery that selects the names and numbers of all customers with ratings equal to the maximum for their city.

```
mysql> SELECT CNUM, CNAME, CITY, RATING FROM customers c1 WHERE
RATING = (SELECT MAX(RATING) FROM customers c2 WHERE c1.CITY =
c2.CITY);
```

+	-----	+	-----	+	-----	+	-----	+
	CNUM		CNAME		CITY		RATING	
+	-----	+	-----	+	-----	+	-----	+
	2001		hoffman		london		100	
	2002		givonanni		rome		200	
	2004		grass		berlin		300	
	2006		clemens		london		100	
	2008		cisneros		san jose		300	
+	-----	+	-----	+	-----	+	-----	+

39. Write a query that totals the orders for each day and places the results in descending order.

```
mysql> SELECT ODATE, SUM(AMT) AS total_amount FROM orders GROUP BY
ODATE ORDER BY total_amount DESC;
```

+	-----	+	-----	+
	ODATE		total_amount	
+	-----	+	-----	+
	1996-06-10		11201.829833984375	
	1996-03-10		8944.590208053589	
	1996-05-10		4723	
	1996-04-10		1788.97998046875	
+	-----	+	-----	+

40. Write a select command that produces the rating followed by the name of each customer in San Jose.

```
mysql> SELECT RATING, CNAME FROM customers WHERE CITY = 'San Jose';
```

+	-----	+	-----	+
	RATING		CNAME	
+	-----	+	-----	+
	200		liu	
	300		cisneros	
+	-----	+	-----	+

41. Find all orders with amounts smaller than any amount for a customer in San Jose.

```
mysql> SELECT * FROM orders WHERE AMT < ( SELECT MIN(AMT)FROM orders
o JOIN customers c ON o.CNUM = c.CNUM WHERE c.CITY = 'San Jose');
```

Empty set (0.00 sec)

42. Find all orders with above average amounts for their customers.

```
mysql> SELECT o.* FROM orders o JOIN (SELECT CNUM, AVG(AMT) AS
avg_amount FROM orders GROUP BY CNUM) avg_orders ON o.CNUM =
avg_orders.CNUM WHERE o.AMT > avg_orders.avg_amount;
```

```
+-----+-----+-----+-----+
| ONUM | AMT   | ODATE   | CNUM | SNUM |
+-----+-----+-----+-----+
| 3006 | 1098.16 | 1996-03-10 | 2008 | 1007 |
| 3009 | 1713.23 | 1996-04-10 | 2002 | 1003 |
| 3011 | 9891.88 | 1996-06-10 | 2006 | 1001 |
+-----+-----+-----+-----+
```

43. Write a query that selects the highest rating in each city.

```
mysql> SELECT CITY, MAX(RATING) AS highest_rating FROM customers GROUP
BY CITY;
```

```
+-----+-----+
| CITY   | highest_rating |
+-----+-----+
| london |          100 |
| rome   |          200 |
| san jose |          300 |
| berlin |          300 |
+-----+-----+
```

44. Write a query that calculates the amount of the salesperson's commission on each order by a customer with a rating above 100.00.

```
mysql> SELECT o.ONUM, o.AMT, s.SNUM, s.COMM, o.AMT * s.COMM AS
commission_amount FROM orders o JOIN customers c ON o.CNUM = c.CNUM JOIN
salespeople s ON o.SNUM = s.SNUM WHERE c.RATING > 100.00;
```

```
+-----+-----+-----+-----+
| ONUM | AMT   | SNUM | COMM | commission_amount |
+-----+-----+-----+-----+
| 3007 | 75.75 | 1003 | 0.1 | 7.575000112876296 |
| 3009 | 1713.23 | 1003 | 0.1 | 171.3230005997866 |
| 3005 | 5160.45 | 1002 | 0.13 | 670.8585007836809 |
| 3010 | 1309.95 | 1002 | 0.13 | 170.29348740601563 |
| 3001 | 18.69 | 1007 | 0.15 | 2.803500191509727 |
| 3006 | 1098.16 | 1007 | 0.15 | 164.724011672497 |
+-----+-----+-----+-----+
```

45. Count the customers with ratings above San Jose's average.

```
mysql> SELECT COUNT(*) AS customer_count FROM customers WHERE RATING >
( SELECT AVG(RATING) FROM customers WHERE CITY = 'San Jose');
```

customer_count
2

46. Write a query that produces all pairs of salespeople with themselves as well as duplicate rows with the order reversed.

```
mysql> SELECT s1.SNUM AS SNUM1, s2.SNUM AS SNUM2 FROM salespeople s1
JOIN salespeople s2 ON s1.SNUM <= s2.SNUM;
```

SNUM1	SNUM2
1001	1001
1001	1002
1001	1003
1001	1004
1001	1005
1001	1007
1002	1002
1002	1003
1002	1004
1002	1005
1002	1007
1003	1003
1003	1004
1003	1005
1003	1007
1004	1004
1004	1005
1004	1007
1005	1005
1005	1007
1007	1007

47. Find all salespeople that are located in either Barcelona or London.

```
mysql> SELECT DISTINCT s.SNUM, s.SNAME, s.CITY FROM salespeople s
WHERE s.CITY IN ('Barcelona', 'London');
```

SNUM	SNAME	CITY
1001	peel	london
1004	motika	london
1005	fran	london
1007	rifkin	barcelona

48. Find all salespeople with only one customer.

```
mysql> SELECT s.SNUM, s.SNAME FROM salespeople s JOIN customers c ON
s.SNUM = c.SNUM GROUP BY s.SNUM, s.SNAME HAVING COUNT(c.CNUM) = 1;
```

SNUM	SNAME
1003	axelrod
1004	motika
1007	rifkin

49. Write a query that joins the Customer table to itself to find all pairs of customers served by a single salesperson.

```
mysql> SELECT c1.CNUM AS CNUM1, c2.CNUM AS CNUM2, c1.SNUM FROM
customers c1 JOIN customers c2 ON c1.SNUM = c2.SNUM AND c1.CNUM <
c2.CNUM;
```

CNUM1	CNUM2	SNUM
2001	2006	1001
2003	2004	1002

50. Write a query that will give you all orders for more than \$1000.00

```
mysql> SELECT * FROM orders WHERE AMT > 1000.00;
```

ONUM	AMT	ODATE	CNUM	SNUM
3002	1900.1	1996-03-10	2007	1004
3005	5160.45	1996-03-10	2003	1002
3006	1098.16	1996-03-10	2008	1007
3008	4723	1996-05-10	2006	1001
3009	1713.23	1996-04-10	2002	1003
3010	1309.95	1996-06-10	2004	1002
3011	9891.88	1996-06-10	2006	1001

51. Write a query that lists each order number followed by the name of the customer who made that order.

```
mysql> SELECT o.ONUM, c.CNAME FROM orders o JOIN customers c ON o.CNUM = c.CNUM;
```

```
+-----+-----+
| ONUM | CNAME  |
+-----+-----+
| 3003 | hoffman |
| 3007 | givonanni |
| 3009 | givonanni |
| 3005 | liu     |
| 3010 | grass   |
| 3008 | clemens |
| 3011 | clemens |
| 3002 | pareira |
| 3001 | cisneros |
| 3006 | cisneros |
+-----+-----+
```

52. Write 2 queries that select all salespeople (by name and number) who have customers in their cities who they do not service, one using a join and one a correlated subquery. Which solution is more elegant?

```
mysql> SELECT DISTINCT s.SNUM, s.SNAME FROM salespeople s JOIN customers c ON c.CITY = s.CITY LEFT JOIN orders o ON s.SNUM = o.SNUM AND c.CNUM = o.CNUM WHERE o.SNUM IS NULL;
```

```
+-----+-----+
| SNUM | SNAME |
+-----+-----+
| 1005 | fran  |
| 1004 | motika |
| 1002 | serres |
+-----+-----+
```

53. Write a query that selects all customers whose ratings are equal to or greater than ANY (in the SQL sense) of Serres'?

```
mysql> SELECT * FROM customers WHERE RATING >= ANY ( SELECT RATING
FROM customers WHERE CNAME = 'Giovanni');
Empty set (0.00 sec)
```

54. Write 2 queries that will produce all orders taken on October 3 or October 4.

```
mysql> SELECT * FROM orders WHERE ODATE IN ('1996-03-10', '1996-04- 10');
```

```
+-----+-----+-----+-----+
| ONUM | AMT   | ODATE   | CNUM | SNUM |
+-----+-----+-----+-----+

```

3001	18.69	1996-03-10	2008	1007
3002	1900.1	1996-03-10	2007	1004
3003	767.19	1996-03-10	2001	1001
3005	5160.45	1996-03-10	2003	1002
3006	1098.16	1996-03-10	2008	1007
+-----+-----+-----+-----+-----+				

55. Write a query that produces all pairs of orders by a given customer. Name that customer and eliminate duplicates.

```
mysql> SELECT o1.ONUM AS ONUM1, o2.ONUM AS ONUM2 FROM orders o1
JOIN orders o2 ON o1.CNUM = o2.CNUM AND o1.ONUM < o2.ONUM WHERE
o1.CNUM = (SELECT CNUM FROM customers WHERE CNAME = 'Giovanni');
```

56. Find only those customers whose ratings are higher than every customer in Rome.

```
mysql> SELECT * FROM customers WHERE RATING > ALL ( SELECT RATING
FROM customers WHERE CITY = 'Rome');
+-----+-----+-----+-----+-----+
| cnum | cname | city | rating | snum |
+-----+-----+-----+-----+-----+
| 2004 | grass | berlin | 300 | 1002 |
| 2008 | cisneros | san jose | 300 | 1007 |
+-----+-----+-----+-----+-----+
```

57. Write a query on the Customers table whose output will exclude all customers with a rating <= 100.00, unless they are located in Rome.

```
mysql> SELECT * FROM customers WHERE RATING > 100.00 OR CITY = 'Rome';
+-----+-----+-----+-----+-----+
| cnum | cname | city | rating | snum |
+-----+-----+-----+-----+-----+
| 2002 | givonanni | rome | 200 | 1003 |
| 2003 | liu | san jose | 200 | 1002 |
| 2004 | grass | berlin | 300 | 1002 |
| 2007 | pareira | rome | 100 | 1004 |
| 2008 | cisneros | san jose | 300 | 1007 |
+-----+-----+-----+-----+-----+
```

58. Find all rows from the Customers table for which the salesperson number is 1001.

```
mysql> SELECT * FROM customers WHERE SNUM = 1001;
+-----+-----+-----+-----+-----+
| cnum | cname | city | rating | snum |
+-----+-----+-----+-----+-----+
```

2001	hoffman	london	100	1001
2006	clemens	london	100	1001

59. Find the total amount in Orders for each salesperson for whom this total is greater than the amount of the largest order in the table.

```
mysql> SELECT s.SNUM, SUM(o.AMT) AS total_amount FROM orders o JOIN
salespeople s ON o.SNUM = s.SNUM GROUP BY s.SNUM HAVING SUM(o.AMT) > (
SELECT MAX(AMT)FROM orders);
```

SNUM	total_amount
1001	15382.069885253906

60. Write a query that selects all orders save those with zeroes or NULLs in the amount field.

```
mysql> SELECT *FROM orders WHERE AMT IS NOT NULL AND AMT > 0;
```

ONUM	AMT	ODATE	CNUM	SNUM
3001	18.69	1996-03-10	2008	1007
3002	1900.1	1996-03-10	2007	1004
3003	767.19	1996-03-10	2001	1001
3005	5160.45	1996-03-10	2003	1002
3006	1098.16	1996-03-10	2008	1007
3007	75.75	1996-04-10	2002	1003
3008	4723	1996-05-10	2006	1001
3009	1713.23	1996-04-10	2002	1003
3010	1309.95	1996-06-10	2004	1002
3011	9891.88	1996-06-10	2006	1001

61. Produce all combinations of salespeople and customer names such that the former precedes the latter alphabetically, and the latter has a rating of less than 200.

```
mysql> SELECT s.SNAME AS Salesperson, c.CNAME AS Customer FROM
salespeople s JOIN customers c ON s.SNAME < c.CNAME WHERE c.RATING < 200;
```

Salesperson	Customer
axelrod	pareira
axelrod	clemens
axelrod	hoffman
motika	pareira
fran	pareira

fran	hoffman	
+-----+	+-----+	+

62. List all Salespeople's names and the Commission they have earned.

```
mysql> SELECT SNAME, COMM FROM salespeople;
```

+-----+	+-----+	+
SNAME	COMM	
+-----+	+-----+	+
peel	0.12	
serres	0.13	
axelrod	0.1	
motika	0.11	
fran	0.26	
rifkin	0.15	
+-----+	+-----+	+

63. Write a query that produces the names and cities of all customers with the same rating as Hoffman. Write the query using Hoffman's CNUM rather than his rating, so that it would still be usable if his rating changed.

```
mysql> SELECT c1.CNAME, c1.CITY FROM customers c1 WHERE c1.RATING = (
SELECT c2.RATING FROM customers c2 WHERE c2.CNUM = 2001);
```

+-----+	+-----+	+
CNAME	CITY	
+-----+	+-----+	+
hoffman	london	
clemens	london	
pareira	rome	
+-----+	+-----+	+

64. Find all salespeople for whom there are customers that follow them in alphabetical order.

```
mysql> SELECT DISTINCT s.SNUM, s.SNAME FROM salespeople s JOIN customers
c1 ON s.SNUM = c1.SNUM JOIN customers c2 ON c1.CNAME < c2.CNAME AND
c1.SNUM = c2.SNUM;
```

+-----+	+-----+	+
SNUM	SNAME	
+-----+	+-----+	+
1002	serres	
1001	peel	
+-----+	+-----+	+

65. Write a query that produces the names and ratings of all customers of all who have above average orders.

66. Find the SUM of all purchases from the Orders table.

```
mysql> SELECT SUM(AMT) AS total_purchases FROM orders;
+-----+
| total_purchases |
+-----+
| 26658.400022506714 |
+-----+
```

67. Write a SELECT command that produces the order number, amount and date for all rows in the order table.

```
mysql> SELECT ONUM, AMT, ODATE FROM orders;
+-----+-----+-----+
| ONUM | AMT   | ODATE   |
+-----+-----+-----+
| 3001 | 18.69 | 1996-03-10 |
| 3002 | 1900.1 | 1996-03-10 |
| 3003 | 767.19 | 1996-03-10 |
| 3005 | 5160.45 | 1996-03-10 |
| 3006 | 1098.16 | 1996-03-10 |
| 3007 | 75.75 | 1996-04-10 |
| 3008 | 4723 | 1996-05-10 |
| 3009 | 1713.23 | 1996-04-10 |
| 3010 | 1309.95 | 1996-06-10 |
| 3011 | 9891.88 | 1996-06-10 |
+-----+-----+-----+
```

68. Count the number of nonNULL rating fields in the Customers table (including repeats).

```
mysql> SELECT COUNT(RATING) AS non_null_ratings FROM customers WHERE
RATING IS NOT NULL;
+-----+
| non_null_ratings |
+-----+
| 7 |
+-----+
```

69. Write a query that gives the names of both the salesperson and the customer for each order after the order number.

```
mysql> SELECT o.ONUM, s.SNAME AS Salesperson, c.CNAME AS Customer FROM
orders o JOIN salespeople s ON o.SNUM = s.SNUM JOIN customers c ON o.CNUM =
c.CNUM;
```

ONUM	Salesperson	Customer
3003	peel	hoffman
3007	axelrod	givonanni
3009	axelrod	givonanni
3005	serres	liu
3010	serres	grass
3008	peel	clemens
3011	peel	clemens
3002	motika	pareira
3001	rifkin	cisneros
3006	rifkin	cisneros

70. List the commissions of all salespeople servicing customers in London.

```
mysql> SELECT DISTINCT s.SNAME, s.COMM FROM salespeople s JOIN customers
c ON s.SNUM = c.SNUM WHERE c.CITY = 'London';
```

SNAME	COMM
peel	0.12

71. Write a query using ANY or ALL that will find all salespeople who have no customers located in their city.

```
mysql> SELECT s.SNUM, s.SNAME FROM salespeople s WHERE NOT EXISTS
(SELECT 1 FROM customers c WHERE c.CITY = s.CITY AND c.SNUM = s.SNUM);
```

SNUM	SNAME
1003	axelrod
1004	motika
1005	fran
1007	rifkin

72. Write a query using the EXISTS operator that selects all salespeople with customers located in their cities who are not assigned to them.

```
mysql> SELECT s.SNUM, s.SNAME FROM salespeople s WHERE EXISTS (SELECT
1 FROM customers c WHERE c.CITY = s.CITY AND c.SNUM != s.SNUM );
```

SNUM	SNAME
1002	serres

1004	motika
1005	fran
+-----+	-----+

73. Write a query that selects all customers serviced by Peel or Motika. (Hint : The SNUM field relates the two tables to one another.)

```
mysql> SELECT DISTINCT c.* FROM customers c JOIN salespeople s ON c.SNUM = s.SNUM WHERE s.SNAME IN ('Peel', 'Motika');
```

+-----+	-----+	-----+	-----+	-----+
cnum	cname	city	rating	snum
+-----+	-----+	-----+	-----+	-----+
2001	hoffman	london	100	1001
2006	clemens	london	100	1001
2007	pareira	rome	100	1004
+-----+	-----+	-----+	-----+	-----+

74. Count the number of salespeople registering orders for each day. (If a salesperson has more than one order on a given day, he or she should be counted only once.)

```
mysql> SELECT ODATE, COUNT(DISTINCT SNUM) AS salesperson_count FROM orders GROUP BY ODATE;
```

+-----+	-----+
ODATE	salesperson_count
+-----+	-----+
1996-03-10	4
1996-04-10	1
1996-05-10	1
1996-06-10	2
+-----+	-----+

75. Find all orders attributed to salespeople in London.

```
mysql> SELECT o.* FROM orders o JOIN salespeople s ON o.SNUM = s.SNUM WHERE s.CITY = 'London';
```

+-----+	-----+	-----+	-----+	-----+
ONUM	AMT	ODATE	CNUM	SNUM
+-----+	-----+	-----+	-----+	-----+
3003	767.19	1996-03-10	2001	1001
3008	4723	1996-05-10	2006	1001
3011	9891.88	1996-06-10	2006	1001
3002	1900.1	1996-03-10	2007	1004
+-----+	-----+	-----+	-----+	-----+

76. Find all orders by customers not located in the same cities as their salespeople.

```
mysql> SELECT o.* FROM orders o JOIN customers c ON o.CNUM = c.CNUM JOIN
salespeople s ON o.SNUM = s.SNUM WHERE c.CITY != s.CITY;
```

ONUM	AMT	ODATE	CNUM	SNUM
3007	75.75	1996-04-10	2002	1003
3009	1713.23	1996-04-10	2002	1003
3010	1309.95	1996-06-10	2004	1002
3002	1900.1	1996-03-10	2007	1004
3001	18.69	1996-03-10	2008	1007
3006	1098.16	1996-03-10	2008	1007

77. Find all salespeople who have customers with more than one current order.

```
mysql> SELECT DISTINCT s.SNUM, s.SNAME FROM salespeople s JOIN customers
c ON s.SNUM = c.SNUM JOIN orders o ON c.CNUM = o.CNUM GROUP BY
s.SNUM, s.SNAME HAVING COUNT(o.ONUM) > 1;
```

SNUM	SNAME
1001	peel
1002	serres
1003	axelrod
1007	rifkin

78. Write a query that extracts from the Customers table every customer assigned to a salesperson who currently has at least one other customer (besides the customer being selected) with orders in the Orders table.

```
mysql> SELECT DISTINCT c.* FROM customers c JOIN salespeople s ON c.SNUM =
s.SNUM WHERE EXISTS (SELECT 1 FROM customers c2 JOIN orders o ON
c2.CNUM = o.CNUM WHERE c2.SNUM = s.SNUM AND c2.CNUM != c.CNUM);
```

cnum	cname	city	rating	snum
2001	hoffman	london	100	1001
2003	liu	san jose	200	1002
2004	grass	berlin	300	1002
2006	clemens	london	100	1001

79. Write a query that selects all customers whose names begin with 'C'.

```
mysql> SELECT * FROM customers WHERE CNAME LIKE 'C%';
```

cnum	cname	city	rating	snum
2006	clemens	london	100	1001
2008	cisneros	san jose	300	1007

80. Write a query on the Customers table that will find the highest rating in each city. Put the output in this form : for the city (city) the highest rating is : (rating).

```
mysql> SELECT CITY, MAX(RATING) AS highest_rating FROM customers GROUP BY CITY;
```

CITY	highest_rating
london	100
rome	200
san jose	300
berlin	300

81. Write a query that will produce the SNUM values of all salespeople with orders currently in the Orders table (without any repeats).

```
mysql> SELECT DISTINCT SNUM FROM orders;
```

SNUM
1001
1002
1003
1004
1007

82. Write a query that lists customers in descending order of rating. Output the rating field first, followed by the customer's names and numbers.

```
mysql> SELECT RATING, CNAME, CNUM FROM customers ORDER BY RATING DESC;
```

RATING	CNAME	CNUM
300	grass	2004
300	cisneros	2008
200	givonanni	2002
200	liu	2003
100	hoffman	2001

100	clemens	2006
100	pareira	2007

83. Find the average commission for salespeople in London.

```
mysql> SELECT AVG(COMM) AS average_commission FROM salespeople WHERE CITY = 'London';
```

average_commission
0.16333332906166712

84. Find all orders credited to the same salesperson who services Hoffman (CNUM 2001).

```
mysql> SELECT o.* FROM orders o WHERE o.SNUM = ( SELECT SNUM FROM customers WHERE CNUM = 2001);
```

ONUM	AMT	ODATE	CNUM	SNUM
3003	767.19	1996-03-10	2001	1001
3008	4723	1996-05-10	2006	1001
3011	9891.88	1996-06-10	2006	1001

85. Find all salespeople whose commission is in between 0.10 and 0.12 (both inclusive).

```
mysql> SELECT * FROM salespeople WHERE COMM BETWEEN 0.10 AND 0.12;
```

snum	sname	city	comm
1001	peel	london	0.12
1003	axelrod	new york	0.1
1004	motika	london	0.11

86. Write a query that will give you the names and cities of all salespeople in London with a commission above 0.10.

```
mysql> SELECT SNAME, CITY FROM salespeople WHERE CITY = 'London' AND COMM > 0.10;
```

SNAME	CITY
-------	------

```

+-----+-----+
| peel | london |
| motika | london |
| fran | london |
+-----+-----+

```

87. What will be the output from the following query? `SELECT * FROM ORDERS where (amt < 1000 OR NOT (odate = 10/03/1996 AND cnum > 2003));`

```
mysql> SELECT * FROM ORDERS WHERE (AMT < 1000 OR NOT (ODATE = '1996-10-03' AND CNUM > 2003));
```

```

+-----+-----+-----+-----+-----+
| ONUM | AMT   | ODATE   | CNUM | SNUM |
+-----+-----+-----+-----+-----+
| 3001 | 18.69 | 1996-03-10 | 2008 | 1007 |
| 3002 | 1900.1 | 1996-03-10 | 2007 | 1004 |
| 3003 | 767.19 | 1996-03-10 | 2001 | 1001 |
| 3005 | 5160.45 | 1996-03-10 | 2003 | 1002 |
| 3006 | 1098.16 | 1996-03-10 | 2008 | 1007 |
| 3007 | 75.75 | 1996-04-10 | 2002 | 1003 |
| 3008 | 4723 | 1996-05-10 | 2006 | 1001 |
| 3009 | 1713.23 | 1996-04-10 | 2002 | 1003 |
| 3010 | 1309.95 | 1996-06-10 | 2004 | 1002 |
| 3011 | 9891.88 | 1996-06-10 | 2006 | 1001 |
+-----+-----+-----+-----+-----+

```

88. Write a query that selects each customer's smallest order.

```
mysql> SELECT CNUM, MIN(AMT) AS smallest_order FROM orders GROUP BY CNUM;
```

```

+-----+-----+
| CNUM | smallest_order |
+-----+-----+
| 2001 | 767.19 |
| 2002 | 75.75 |
| 2003 | 5160.45 |
| 2004 | 1309.95 |
| 2006 | 4723 |
| 2007 | 1900.1 |
| 2008 | 18.69 |
+-----+-----+

```

89. Write a query that selects the first customer in alphabetical order whose name begins with G.

```
mysql> SELECT CNAME, CNUM FROM customers WHERE CNAME LIKE 'G%' ORDER BY CNAME LIMIT 1;
```

```

+-----+-----+

```

CNAME	CNUM
givonanni	2002

90. Write a query that counts the number of different nonNULL city values in the Customers table.

```
mysql> SELECT COUNT(DISTINCT CITY) AS unique_non_null_cities FROM customers WHERE CITY IS NOT NULL;
```

unique_non_null_cities
4

91. Find the average amount from the Orders table.

```
mysql> SELECT AVG(AMT) AS average_amount FROM orders;
```

average_amount
2665.840002250671

92. What would be the output from the following query? SELECT * FROM ORDERS WHERE NOT (odate = 10/03/96 OR snum > 1006) AND amt >= 1500);

```
mysql> SELECT * FROM ORDERS WHERE NOT (ODATE = '1996-03-10' OR SNUM > 1006) AND AMT >= 1500;
```

ONUM	AMT	ODATE	CNUM	SNUM
3008	4723	1996-05-10	2006	1001
3011	9891.88	1996-06-10	2006	1001
3009	1713.23	1996-04-10	2002	1003

93. Find all customers who are not located in San Jose and whose rating is above 200.

```
mysql> SELECT * FROM customers WHERE CITY <> 'San Jose' AND RATING > 200;
```

cnum	cname	city	rating	snum
2004	grass	berlin	300	1002

94. Give a simpler way to write this query : SELECT snum, sname city, comm FROM salespeople WHERE (comm > + 0.12 OR comm < 0.14);

```
mysql> SELECT SNUM, SNAME, CITY, COMM FROM salespeople WHERE COMM
BETWEEN 0.12 AND 0.14;
```

SNUM	SNAME	CITY	COMM
1002	serres	san jose	0.13

95. Evaluate the following query : SELECT * FROM orders WHERE NOT ((odate = 10/03/96 AND snum > 1002) OR amt > 2000.00);

```
mysql> SELECT * FROM orders WHERE NOT ((ODATE = '1996-10-03' AND SNUM >
1002 ) OR AMT > 2000.00);
```

ONUM	AMT	ODATE	CNUM	SNUM
3001	18.69	1996-03-10	2008	1007
3002	1900.1	1996-03-10	2007	1004
3003	767.19	1996-03-10	2001	1001
3006	1098.16	1996-03-10	2008	1007
3007	75.75	1996-04-10	2002	1003
3009	1713.23	1996-04-10	2002	1003
3010	1309.95	1996-06-10	2004	1002

96. Which salespersons attend to customers not in the city they have been assigned to?

```
mysql> SELECT DISTINCT s.SNUM, s.SNAME FROM salespeople s JOIN
-> customers c ON s.SNUM = c.SNUM WHERE s.CITY <> c.CITY;
```

SNUM	SNAME
1003	axelrod
1002	serres
1004	motika
1007	rifkin

97. Which salespeople get commission greater than 0.11 are serving customers rated less than 250?

```
mysql> SELECT DISTINCT s.SNUM, s.SNAME FROM salespeople s JOIN customers c
ON s.SNUM = c.SNUM WHERE s.COMM > 0.11 AND c.RATING < 250;
```

```
+-----+-----+
| SNUM | SNAME |
+-----+-----+
| 1001 | peel  |
| 1002 | serres|
+-----+-----+
```

98. Which salespeople have been assigned to the same city but get different commission percentages?

```
mysql> SELECT s1.CITY, s1.COMM AS COMM1, s2.COMM AS COMM2 FROM
salespeople s1 JOIN salespeople s2 ON s1.CITY = s2.CITY AND s1.SNUM <> s2.SNUM
WHERE s1.COMM <> s2.COMM
```

```
-> ;
+-----+-----+-----+
| CITY  | COMM1 | COMM2 |
+-----+-----+-----+
| london | 0.26 | 0.12 |
| london | 0.11 | 0.12 |
| london | 0.26 | 0.11 |
| london | 0.12 | 0.11 |
| london | 0.11 | 0.26 |
| london | 0.12 | 0.26 |
+-----+-----+-----+
```

99. Which salesperson has earned the most by way of commission?

```
mysql> SELECT SNUM, SNAME, COMM FROM salespeople ORDER BY COMM
DESC LIMIT 1;
```

```
+-----+-----+-----+
| SNUM | SNAME | COMM |
+-----+-----+-----+
| 1005 | fran  | 0.26 |
+-----+-----+-----+
```

100. Does the customer who has placed the maximum number of orders have the maximum rating?

```
mysql> WITH MaxOrderCustomer AS (SELECT CNUM FROM orders GROUP BY
CNUM ORDER BY COUNT(ONUM) DESC LIMIT 1), MaxRatingCustomer AS
(SELECT CNUM, RATING FROM customers WHERE CNUM = (SELECT CNUM
FROM MaxOrderCustomer)) SELECT MAX(RATING) AS max_rating FROM customers
WHERE CNUM = (SELECT CNUM FROM MaxOrderCustomer);
```

```
+-----+
| max_rating |
+-----+
```

200

101.Has the customer who has spent the largest amount of money been given the highest rating?

```
mysql> WITH MaxAmountCustomer AS (SELECT CNUM FROM orders
-> GROUP BY CNUM ORDER BY SUM(AMT) DESC LIMIT 1 ),MaxRating AS
-> (SELECT RATING FROM customers WHERE CNUM = (SELECT CNUM
-> FROM MaxAmountCustomer))SELECT MAX(RATING) AS highest_rating
-> FROM customers WHERE CNUM = (SELECT CNUM FROM
-> MaxAmountCustomer);
```

highest_rating
100

102.List all customers in descending order of customer rating.

```
mysql> SELECT CNAME, RATING FROM customers ORDER BY RATING DESC;
```

CNAME	RATING
grass	300
cisneros	300
givonanni	200
liu	200
hoffman	100
clemens	100
pareira	100

103.On which days has Hoffman placed orders?

```
mysql> SELECT DISTINCT ODATE FROM orders WHERE CNUM = 2001;
```

ODATE
1996-03-10

104.Do all salespeople have different commissions?

```
mysql> SELECT COUNT(DISTINCT COMM) AS unique_commissions,
COUNT(COMM) AS total_commissions FROM salespeople;
```

unique_commissions	total_commissions
6	6

105.Which salespeople have no orders between 10/03/1996 and 10/05/1996?

```
mysql> SELECT s.SNUM, s.SNAME FROM salespeople s WHERE NOT EXISTS
(SELECT 1 FROM orders o WHERE o.SNUM = s.SNUM AND o.ODATE BETWEEN
'1996-10-03' AND '1996-10-05');
```

SNUM	SNAME
1001	peel
1002	serres
1003	axelrod
1004	motika
1005	fran
1007	rifkin

106.How many salespersons have succeeded in getting orders?

```
mysql> SELECT COUNT(DISTINCT SNUM) AS number_of_salespeople FROM
orders;
```

number_of_salespeople
5

107.How many customers have placed orders?

```
mysql> SELECT COUNT(DISTINCT CNUM) AS number_of_customers FROM orders;
```

number_of_customers

```
|          7 |
+-----+
```

108. On which date has each salesperson booked an order of maximum value?

```
mysql> WITH MaxOrder AS (SELECT SNUM, MAX(AMT) AS max_amount FROM
orders GROUP BY SNUM) SELECT o.SNUM, o.ODATE FROM orders o JOIN
MaxOrder mo ON o.SNUM = mo.SNUM AND o.AMT = mo.max_amount;
```

```
+-----+-----+
| SNUM | ODATE   |
+-----+-----+
| 1004 | 1996-03-10 |
| 1002 | 1996-03-10 |
| 1007 | 1996-03-10 |
| 1003 | 1996-04-10 |
| 1001 | 1996-06-10 |
+-----+-----+
```

109. Who is the most successful salesperson?

```
mysql> SELECT SNUM, SNAME FROM salespeople ORDER BY COMM DESC
LIMIT 1;
```

```
+-----+-----+
| SNUM | SNAME |
+-----+-----+
| 1005 | fran  |
+-----+-----+
```

110. Who is the worst customer with respect to the company?

```
mysql> SELECT CNAME, CNUM FROM customers ORDER BY RATING ASC
LIMIT 1;
```

```
+-----+-----+
| CNAME | CNUM |
+-----+-----+
| hoffman | 2001 |
+-----+-----+
```

111. Are all customers not having placed orders greater than 200 totally been serviced by salespersons Peel or Serres?

```
mysql> SELECT COUNT(*) AS unmatched_customers FROM customers c WHERE
RATING > 200 AND CNUM NOT IN (SELECT CNUM FROM orders) AND SNUM
NOT IN (SELECT SNUM FROM salespeople WHERE SNAME IN ('Peel', 'Serres'));
```

```
+-----+
| unmatched_customers |
+-----+
|          0 |
+-----+
```

112. Which customers have the same rating?

```
mysql> SELECT RATING FROM customers GROUP BY RATING HAVING  
COUNT(*) > 1;
```

```
+-----+  
| RATING |  
+-----+  
| 100 |  
| 200 |  
| 300 |  
+-----+
```

113. Find all orders greater than the average for October 4th.

```
mysql> WITH October4Avg AS ( SELECT AVG(AMT) AS avg_amount FROM orders  
WHERE ODATE = '1996-04-10') SELECT * FROM orders WHERE ODATE =  
'1996-04-10' AND AMT > (SELECT avg_amount FROM October4Avg)  
-> ;
```

```
+-----+-----+-----+-----+-----+  
| ONUM | AMT | ODATE | CNUM | SNUM |  
+-----+-----+-----+-----+-----+  
| 3009 | 1713.23 | 1996-04-10 | 2002 | 1003 |  
+-----+-----+-----+-----+-----+
```

114. Which customers have above average orders?

```
mysql> WITH CustomerAvgOrder AS (SELECT CNUM, AVG(AMT) AS  
avg_order_amount FROM orders GROUP BY CNUM), OverallAvg AS ( SELECT  
AVG(avg_order_amount) AS overall_avg FROM CustomerAvgOrder) SELECT  
c.CNAME, c.CNUM FROM customers c JOIN CustomerAvgOrder cao ON c.CNUM =  
cao.CNUM WHERE cao.avg_order_amount > (SELECT overall_avg FROM  
OverallAvg);
```

```
+-----+-----+  
| CNAME | CNUM |  
+-----+-----+  
| liu | 2003 |  
| clemens | 2006 |  
+-----+-----+
```

115. List all customers with ratings above San Jose's average.

```
mysql> WITH SanJoseAvg AS (SELECT AVG(RATING) AS avg_rating FROM  
customers WHERE CITY = 'San Jose') SELECT * FROM customers WHERE RATING  
> (SELECT avg_rating FROM SanJoseAvg);
```

```
+-----+-----+-----+-----+-----+  
| cnum | cname | city | rating | snum |  
+-----+-----+-----+-----+-----+  
| 2004 | grass | berlin | 300 | 1002 |  
| 2008 | cisneros | san jose | 300 | 1007 |  
+-----+-----+-----+-----+-----+
```

116. Select the total amount in orders for each salesperson for whom the total is greater than the amount of the largest order in the table.

```
mysql> WITH MaxOrderAmount AS (SELECT MAX(AMT) AS largest_order_amount
FROM orders),SalespersonTotals AS ( SELECT SNUM, SUM(AMT) AS total_amount
FROM orders GROUP BY SNUM) SELECT s.SNUM, s.total_amount FROM
SalespersonTotals s JOIN MaxOrderAmount m ON s.total_amount >
m.largest_order_amount;
+-----+-----+
| SNUM | total_amount |
+-----+-----+
| 1001 | 15382.069885253906 |
+-----+-----+
```

117. Give names and numbers of all salespersons who have more than one customer.

```
mysql> SELECT SNUM, SNAME FROM salespeople WHERE SNUM IN (SELECT
SNUM FROM customers GROUP BY SNUM HAVING COUNT(CNUM) > 1 );
+-----+-----+
| SNUM | SNAME |
+-----+-----+
| 1001 | peel |
| 1002 | serres |
+-----+-----+
```

118. Select all salespersons by name and number who have customers in their city whom they don't service.

```
mysql> SELECT DISTINCT s.SNUM, s.SNAME FROM salespeople s JOIN customers
c ON s.CITY = c.CITY WHERE s.SNUM <> c.SNUM;
+-----+-----+
| SNUM | SNAME |
+-----+-----+
| 1005 | fran |
| 1004 | motika |
| 1002 | serres |
+-----+-----+
```

119. Which customers' rating should be lowered?

```
mysql> WITH AverageRating AS ( SELECT AVG(RATING) AS avg_rating FROM
customers) SELECT CNAME, RATING FROM customers WHERE RATING >
(SELECT avg_rating * 2 FROM AverageRating);
Empty set (0.00 sec)
```

120. Is there a case for assigning a salesperson to Berlin? mysql>

```
mysql> SELECT DISTINCT s.SNUM, s.SNAME FROM salespeople s WHERE NOT
EXISTS (SELECT 1 FROM customers c WHERE c.CITY = 'Berlin' AND c.SNUM =
s.SNUM );
```

SNUM	SNAME
1001	peel
1003	axelrod
1004	motika
1005	fran
1007	rifkin

121.Is there any evidence linking the performance of a salesperson to the commission that he or she is being paid?

```
mysql> SELECT s.SNUM, s.SNAME, AVG(o.AMT) AS avg_order_amount, s.COMM
FROM salespeople s JOIN orders o ON s.SNUM = o.SNUM GROUP BY s.SNUM,
s.SNAME, s.COMM ORDER BY s.COMM;
```

SNUM	SNAME	avg_order_amount	COMM
1003	axelrod	894.489990234375	0.1
1004	motika	1900.0999755859375	0.11
1001	peel	5127.356628417969	0.12
1002	serres	3235.2000732421875	0.13
1007	rifkin	558.4250173568726	0.15

122.Does the total amount in orders by customer in Rome and London exceed the commission paid to salespersons in London and New York by more than 5 times?

```
mysql> WITH TotalOrdersRomeLondon AS (SELECT SUM(o.AMT) AS total_orders
FROM orders o JOIN customers c ON o.CNUM = c.CNUM WHERE c.CITY IN
('Rome', 'London')),TotalCommissionLondonNewYork AS (SELECT SUM(s.COMM)
AS total_commission FROM salespeople s WHERE s.CITY IN ('London', 'New York'))
SELECT (SELECT total_orders FROM TotalOrdersRomeLondon) > 5 * (SELECT
total_commission FROM TotalCommissionLondonNewYork) AS ExceedsFiveTimes;
```

ExceedsFiveTimes
1

123. Which is the date, order number, amt and city for each salesperson (by name) for the maximum order he has obtained?

```
mysql> WITH MaxOrderPerSalesperson AS (SELECT SNUM, MAX(AMT) AS
max_amount FROM orders GROUP BY SNUM) SELECT s.SNAME, o.ODATE,
o.ONUM, o.AMT, c.CITY FROM orders o JOIN MaxOrderPerSalesperson mo ON
o.SNUM = mo.SNUM AND o.AMT = mo.max_amount JOIN salespeople s ON
o.SNUM = s.SNUM JOIN customers c ON o.CNUM = c.CNUM;
```

SNAME	ODATE	ONUM	AMT	CITY
peel	1996-06-10	3011	9891.88	london
serres	1996-03-10	3005	5160.45	san jose
axelrod	1996-04-10	3009	1713.23	rome
motika	1996-03-10	3002	1900.1	rome
rifkin	1996-03-10	3006	1098.16	san jose

124. Which salesperson(s) should be fired?

```
mysql> SELECT s.SNUM, s.SNAME FROM salespeople s LEFT JOIN orders
-> o ON s.SNUM = o.SNUM GROUP BY s.SNUM, s.SNAME HAVING
-> SUM(o.AMT) IS NULL AND AVG(s.COMM) < 0.05;
Empty set (0.00 sec)
```

125. What is the total income for the company?

```
mysql> SELECT SUM(AMT) AS total_income FROM orders;
+-----+
| total_income |
+-----+
| 26658.400022506714 |
+-----+
```